Unit

37 Factors affecting the rate of a reaction

Unit Key Concepts

- Factors affecting the rate of reaction: concentration, surface area, temperature and catalyst
- Relationship between reaction rate and effective collisions
- Applications of catalysts and enzymes

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

37.1 Factors affecting the rate of a reaction

There are several factors that affect the rate of a reaction:

- the concentration of a reactant;
- the surface area of a solid reactant;
- the temperature; and
- the presence of a catalyst.

We are going to consider each of these factors in turn.

Assessment Form

Assessment forms on these activities are available for download from *Jing Kung Chemistry Website*.

37.1 **Assessment Form**

Investigating the effect of varying the concentration of hydrochloric acid on the rate of its reaction with magnesium.

37.2 **Assessment Form**

Investigating the effect of varying the concentration of hydroxide ion on the rate of its reaction with phenolphthalein.

37.2 Studying the effect of change in concentration of a reactant on the rate of a reaction

Consider the reaction between magnesium ribbons of equal length with 1.0 mol dm^{-3} hydrochloric acid and 0.50 mol dm^{-3} hydrochloric acid separately. Each time the same volume of acid (excess) was used.

$$Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$$

Fig. 37.1a shows the curves of volume of hydrogen released plotted against time for the two different concentrations of hydrochloric acid.

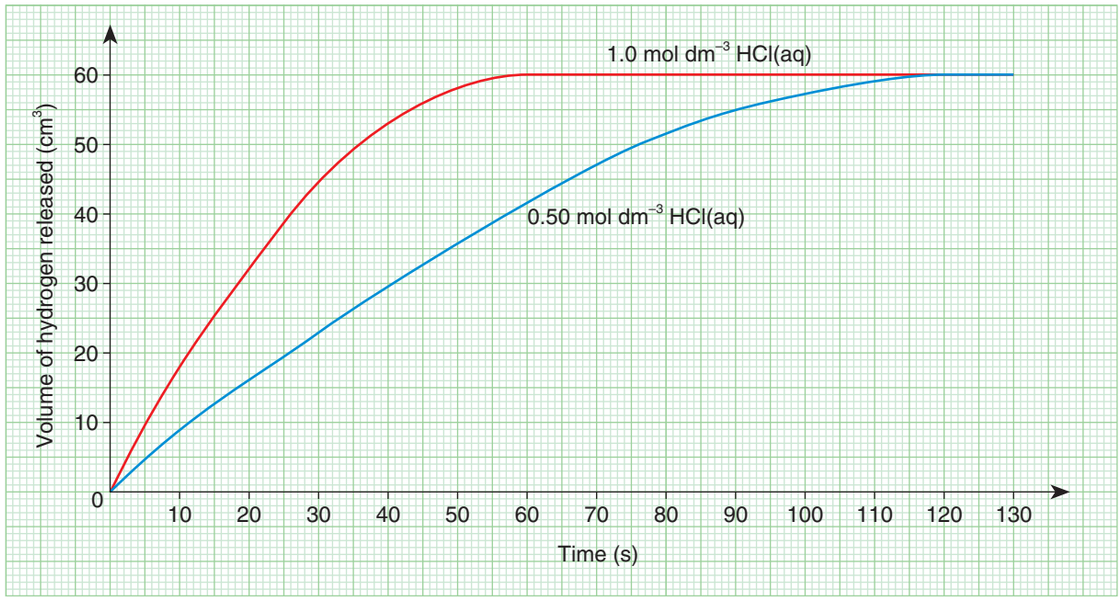


Fig. 37.1a Curves showing the volume of hydrogen released plotted against time for the two different concentrations of hydrochloric acid

N1 Refer to page T16.

The curves highlight the following points: **N1**

CE 2009 Paper 2 Q43
CE 2010 Paper 2 Q44

- At the start of the reaction, the tangent to the curve for the 1.0 mol dm^{-3} hydrochloric acid is steeper than that for the 0.50 mol dm^{-3} hydrochloric acid. That means the initial rate of reaction is faster when 1.0 mol dm^{-3} hydrochloric acid is used.
- The 1.0 mol dm^{-3} hydrochloric acid takes less time to complete the reaction.
- The same total volume of hydrogen is formed. This is because magnesium is the limiting reactant.



Fig. 37.1b The rate of reaction increases (i.e. more fizzing seen) as the concentration of the acid increases (from left to right)

✓ In most cases, the rate of a reaction increases when the concentration of a reactant is increased.

Example 37.1

Q Phenolphthalein is a commonly used indicator in acid-alkali titrations. It is colourless in solutions of pH 8 or lower. We can represent the structure of the colourless form of phenolphthalein as H_2Ph. The addition of hydroxide ions removes the hydrogen atoms from phenolphthalein, producing the pink $Ph^{2-}(aq)$ ions.

$$H_2Ph(aq) + 2OH^-(aq) \rightleftharpoons Ph^{2-}(aq) + 2H_2O(l)$$

If still more $OH^-(aq)$ ions are added, $Ph^{2-}(aq)$ ions would react with the $OH^-(aq)$ ions to form a colourless complex, $PhOH^{3-}(aq)$. The pink colour would fade slowly.

$$Ph^{2-}(aq) + OH^-(aq) \rightleftharpoons PhOH^{3-}(aq)$$

The following samples are used in an experiment to study the rate of the reaction between $Ph^{2-}(aq)$ and $OH^-(aq)$ ions.

Sample	Composition of sample			
	Volume of 0.3 mol dm^{-3} NaOH(aq) (cm^3)	Volume of 0.3 mol dm^{-3} NaCl(aq) (cm^3)	Volume of phenolphthalein (drops)	Reaction time (min)
1	2.0	8.0	2	8.0
2	4.0	6.0	2	4.0
3	6.0	4.0	2	2.6
4	8.0	2.0	2	2.0

a) Expain why

i) the total volume of each sample is kept the same;

ii) the same number of drops of phenolphthalein is used in each sample.

b) Suggest how the reaction time could be measured.

c) Describe how the rate of the reaction varies with the concentration of $OH^-(aq)$ ions.

A a) i) The concentration of $OH^-(aq)$ ions in the sample is directly proportional to the volume of NaOH(aq) used.

ii) To ensure that the only variable is the change in the concentration of $OH^-(aq)$ ions in the sample.

b) Measure the time required for the phenolphthalein to decolourize.

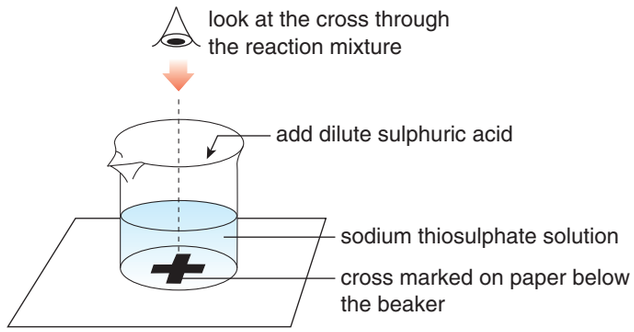
c) If the reaction is fast, the time for the phenolphthalein to decolourize will be short. If the reaction is slow, the time will be long.

Consider samples 1 and 2. When the concentration of $OH^-(aq)$ ions doubles, the reaction time decreases to one-half. Thus, the rate of the reaction between $Ph^{2-}(aq)$ and $OH^-(aq)$ ions increases with the concentration of $OH^-(aq)$ ions.

Practice 37.1

The set-up shown below was used to study how the concentration of sodium thiosulphate solution affects the rate of the following reaction:

$Na_2S_2O_3(aq) + H_2SO_4(aq) \longrightarrow Na_2SO_4(aq) + SO_2(g) + H_2O(l) + S(s)$



The following samples were used in the experiment. The time *t* required for the cross to disappear when viewed from above was recorded for each sample.

Sample	Composition of sample			
	Volume of 0.1 mol dm^{-3} $Na_2S_2O_3$ (cm^3)	Volume of water (cm^3)	Volume of 1 mol dm^{-3} H_2SO_4 (cm^3)	Time *t* (s)
1	4.0	0.0	4.0	37
2	3.0	1.0	4.0	50
3	2.0	2.0	4.0	76
4	1.0	3.0	4.0	154

a) Describe how the rate of the reaction can be obtained from the experimental results.

b) Explain why

i) water was used in samples 2 to 4;

ii) the same volume of acid was used in each sample.

c) Deduce how the change in concentration of sodium thiosulphate affects the rate of the reaction.

Refer to page T17.

Practice 37.1

a) Rate of the reaction $\propto \frac{1}{t}$

b) i) To keep the total volume of each sample constant. Thus, the concentration of sodium thiosulphate in the sample is directly proportional to the volume of thiosulphate solution used.

ii) To ensure that the only variable is the change in the concentration of sodium thiosulphate in the sample.

37.3 Studying the effect of change in surface area of a solid reactant on the rate of a reaction

Consider the following chemical reactions:

- Cooking — potatoes take a shorter time to cook when they are cut into smaller pieces;
- Combustion — small pieces of stick burn more quickly than a large block of wood (Fig. 37.2).

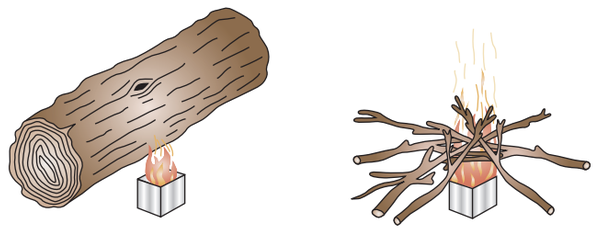


Fig. 37.2 The surface area of the wood exposed to the air makes all the difference to the rate of the combustion reaction

The reactions occur more quickly because the **surface area** of the solid has been increased.

A fixed mass of solid in a single block has a smaller surface area than the same mass of solid cut into smaller pieces (Fig. 37.3).

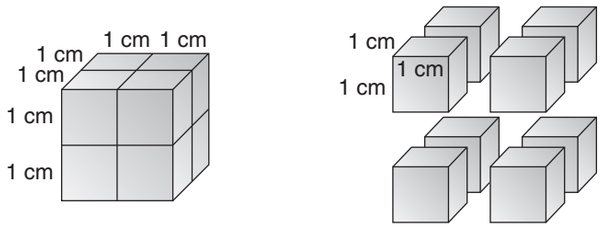


Fig. 37.3 Eight 1 cm cubes have the same mass but twice the surface area of one 2 cm cube

Consider the reaction between equal masses of large marble chips and crushed marble chips with hydrochloric acid. Each time the same volume of 1.0 mol dm^{-3} hydrochloric acid (excess) is used.

$$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$$

Fig. 37.4 shows the curves of loss in mass of the reaction mixture plotted against time for the two types of marble chips. The loss in mass of each reaction mixture equals the mass of carbon dioxide released.

Assessment Form
An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

37.3
Investigating the effect of varying the surface area of marble chips on the rate of their reaction with dilute hydrochloric acid.

surface area 表面面積

DSE 2013 Paper 1A Q25

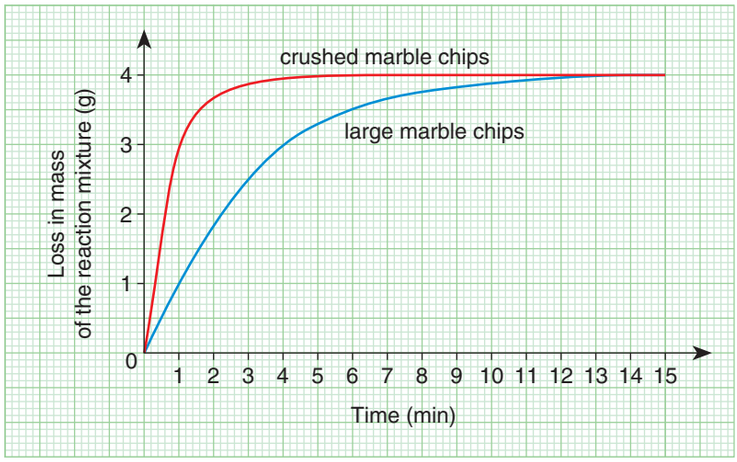


Fig. 37.4 Curves showing the loss in mass of the reaction mixture plotted against time for the two types of marble chips

The curves highlight the following points:

CE 2005 Paper 2 Q50
DSE 2013 Paper 1B Q11(a)
ASL 2006 Q7(a)

- At the start of the reaction, the tangent to the curve for the crushed marble chips is steeper than that for the large marble chips. That means the initial rate of reaction is higher when crushed marble chips are used.
- The crushed marble chips take less time to complete the reaction.
- There is the same final loss in mass of the reaction mixture. That means the same mass of carbon dioxide is formed in the two reactions. This is because the amount of marble chips limits the mass of carbon dioxide formed.

✓ The rate of a reaction increases when the surface area of a solid reactant is increased. N2

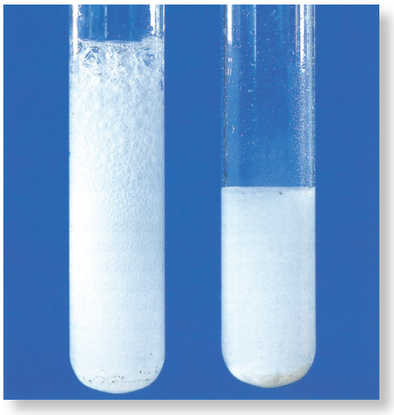


Fig. 37.5 In the reaction with dilute hydrochloric acid, crushed marble chips (left) react faster than large marble chips (right)

N2 Refer to page T17.

Do you know?

Powders in factories

Combustible substances burn easily. Combustible powders such as flour, custard powder and sulphur burn explosively in air. Thus, while these powders are manufactured, care must be taken to prevent their dust from escaping into the air inside the factory building. Sparks or naked flames must also be prohibited inside the building.

Example 37.2

Q A student performed an experiment to investigate the rate of reaction between magnesium and an acid. 2.0 g of magnesium ribbons were added to a conical flask containing 100 cm^3 of 2.0 mol dm^{-3} hydrochloric acid (excess) at 20 °C.

Afterwards the experiment was repeated with each of the following changes. In each case, state and explain whether the expected reaction rate would increase, decrease or remain unchanged.

a) 2.0 g of magnesium powder were used instead of magnesium ribbons. CE 2010 Paper 2 Q44; ASL 2006 Q7(a)

b) 200 cm^3 of 2.0 mol dm^{-3} hydrochloric acid were used instead of 100 cm^3 of the original acid.

c) 100 cm^3 of 2.0 mol dm^{-3} ethanoic acid were used instead of hydrochloric acid. CE 2005 Paper 2 Q34; ASL 2006 Q7(b)

CE 2010 Paper 2 Q44

A a) Increase

There was an increase in the surface area of magnesium.

b) Remain unchanged

The concentration of the hydrochloric acid remained the same.

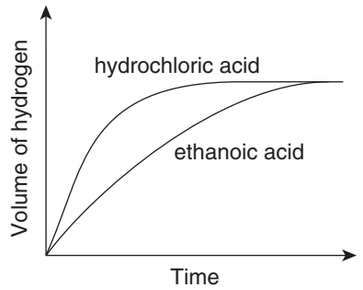
c) Decrease

In the reaction between magnesium and the acids, magnesium reacts with hydrogen ions in the acids.

Ethanoic acid is a weak acid while hydrochloric acid is a strong acid.

2.0 mol dm^{-3} ethanoic acid had a smaller concentration of hydrogen ions than 2.0 mol dm^{-3} hydrochloric acid.

If the concentration and volume of the acids are the same, and the same mass of magnesium is used, the same volume of gas is produced. It is just produced faster in the reaction with hydrochloric acid.



Practice 37.2

To study the rate of reaction of zinc with acids, a student carried out the following experiments at room temperature. Each time the acid was in excess.

Experiment	Reactants mixed	Time required for reaction to complete (s)
I	1 g of zinc powder + 50 cm^3 of 1 mol dm^{-3} HCl(aq)	200
II	1 g of zinc powder + 50 cm^3 of 1 mol dm^{-3} H_2SO_4(aq)	t_1
III	1 g of zinc granules + 50 cm^3 of 1 mol dm^{-3} HCl(aq)	t_2

Continued on next page

a) How could you know when the reactions were complete?
All the zinc was used up and dissolved in the acid.

b) State whether you would expect

i) t_1;

ii) t_2

to be more or less than 200 s respectively. Explain your answer in each case.

CE 2008 Paper 2 Q39
DSE 2012 Paper 1B Q11(b)

Refer to page T18.

37.4 Studying the effect of change in temperature on the rate of a reaction

Consider again the reaction between magnesium ribbon and dilute hydrochloric acid. Carry out the reaction at 15 °C and 25 °C respectively. Each time the same length of magnesium ribbon and same volume of 0.50 mol dm^{-3} hydrochloric acid (excess) are used.

Fig. 37.6 shows the curves of volume of hydrogen released plotted against time for the reactions carried out at 15 °C and 25 °C.

N3 Refer to page T18.

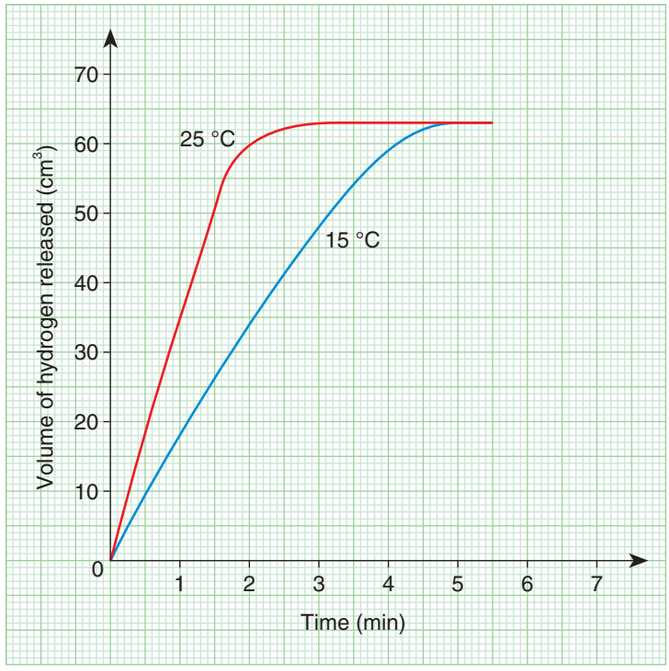


Fig. 37.6 Curves showing the volume of hydrogen released plotted against time for the reaction between magnesium and dilute hydrochloric acid at different temperatures

Assessment Form
Assessment forms on these activities are available for download from *Jing Kung Chemistry Website*.

37.4
Investigating the effect of varying the temperature on the rate of the reaction between sodium thiosulphate solution and dilute sulphuric acid. N3 Assessment Form

37.5
Investigating the effect of varying the temperature on the rate of the reaction between ingredients of Alka Seltzer tablet. Assessment Form

Video
Investigating the effect of varying the temperature on the rate of the reaction between sodium thiosulphate solution and dilute sulphuric acid

Video
Investigating the effect of varying the temperature on the rate of the reaction between ingredients of Alka Seltzer tablet

The curves highlight the following points:

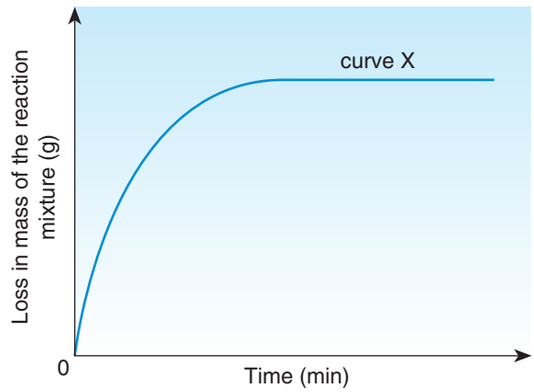
- At the start of the reaction, the tangent to the curve for the reaction at 25 °C is steeper than that at 15 °C. That means the initial rate of reaction is higher when the reaction is carried out at 25 °C.
- The reaction at 25 °C takes less time to complete.
- The same total volume of hydrogen is released.

✓ In most cases, the rate of a reaction increases when the temperature is increased.

Practice 37.3

Refer to page T19.

Some powdered calcium carbonate are allowed to react with excess 0.50 mol dm^{-3} hydrochloric acid in a conical flask. The loss in mass of the reaction mixture is recorded at regular time intervals. The results are represented by curve X.



Sketch next to curve X the curve you expect for each of the following changes made:

a) using a lump of calcium carbonate of the same mass;

b) using a lump of calcium carbonate of lower mass;

c) using the same volume of 1.0 mol dm^{-3} hydrochloric acid;

d) increasing the temperature.

N4 Refer to page T19.

CE 2006 Paper 2 Q44
ASL 2006 Q10(a)(ii)

37.5 Studying how the presence of a catalyst affects the rate of a reaction

✓ A **catalyst** is a substance which alters the rate of a reaction without itself undergoing any permanent chemical changes. N4

The mass of a catalyst remains unchanged at the end of a reaction. There are two types of catalysts: positive catalysts and negative catalysts. A positive catalyst is one that speeds up a reaction. A negative catalyst is one that slows down a reaction.

One way of making oxygen in the school laboratory is by the decomposition of hydrogen peroxide, H_2O_2.

$$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$$ CE 2009 Paper 1 Q10(a)(iii)

Under normal conditions, the decomposition of hydrogen peroxide is very slow. Suppose we want to find out whether manganese(IV) oxide can catalyze the decomposition.

Let us follow the progress of the decomposition of hydrogen peroxide without and with manganese(IV) oxide. Fig. 37.7 shows the set-up for the first experiment — without manganese(IV) oxide. 50 cm^3 of 1.0 mol dm^{-3} hydrogen peroxide solution are allowed to decompose in the suction flask. The pressure inside the flask increases as oxygen gas is formed. Record the pressure inside the flask in the first 5 minutes.

Do you know

Catalysts and the ozone layer

Not until recently, chlorofluorocarbons (CFCs) were still used for a number of applications, such as aerosol propellants. CFCs escape high into the atmosphere and eventually decompose to produce chlorine atoms. These atoms act as catalysts in reactions that bring about the destruction of ozone, O_3.

chlorine atom catalyst

$$O_3(g) + O(g) \longrightarrow 2O_2(g)$$

International agreements have resulted in CFCs being phased out. Unfortunately there is still CFCs remaining in the atmosphere.

37.6

Catalyzing the decomposition of hydrogen peroxide in solution.

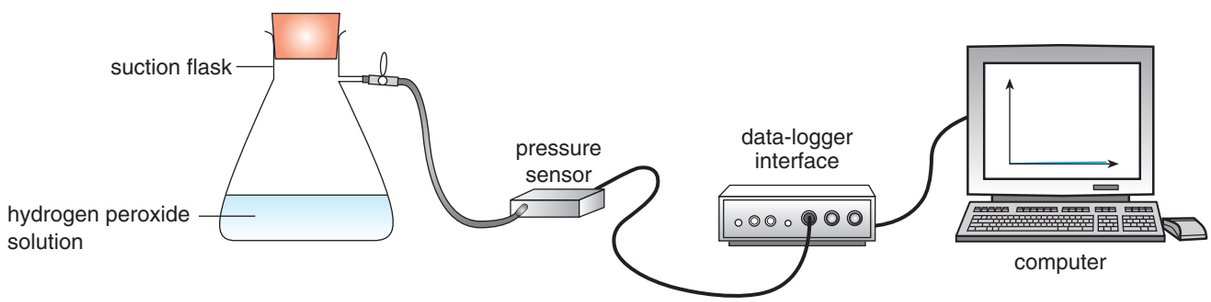


Fig. 37.7 Experimental set-up for following the progress of the decomposition of hydrogen peroxide without manganese(IV) oxide

catalyst 催化劑

Caution: Oxygen gas produced in the reaction will cause the pressure inside the flask to increase.

Fig. 37.8 shows the set-up used for the second experiment — with manganese(IV) oxide. Use the same concentration and volume of hydrogen peroxide solution as before. To start the experiment, tilt the bottle so that the manganese(IV) oxide comes into contact with the hydrogen peroxide solution. Record the pressure inside the suction flask in the first 5 minutes.

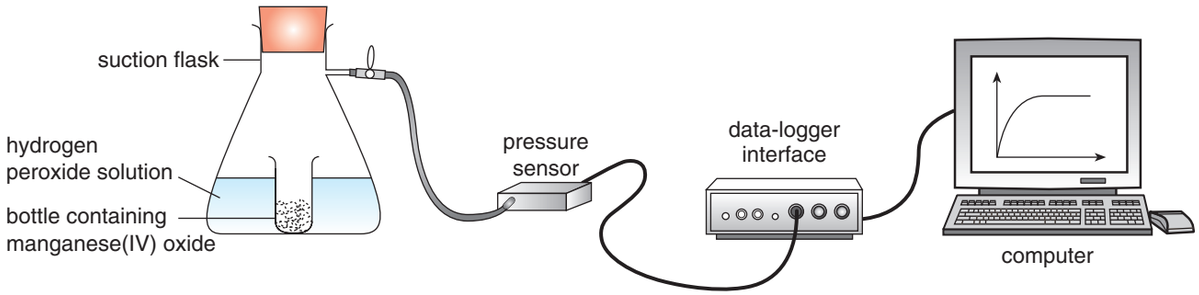


Fig. 37.8 Experimental set-up for following the progress of the decomposition of hydrogen peroxide with manganese(IV) oxide

Reference Website
Refer to the following website for a simulation of the progress of the catalytic decomposition of hydrogen peroxide solution:
http://group.chem.iastate.edu/Greenbowe/sections/projectfolder/flashfiles/kinetics2/rxnRate01.html
(accessed August 2014)

Fig. 37.9 shows the curves of pressure plotted against time for the two experiments.

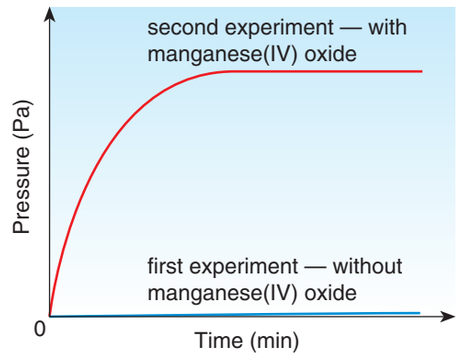


Fig. 37.9 The curves of pressure plotted against time for the two experiments

The curves highlight the following points:

- At the start of the decomposition, the tangent to the curve for the second experiment is steeper than that for the first experiment. That means the initial rate of decomposition is higher when manganese(IV) oxide is present. Manganese(IV) oxide is a positive catalyst for the decomposition (Fig. 37.10).

CE 2009 Paper 1 Q10(a)(i)
DSE 2014 Paper 1A Q25

- When the first experiment is still taking place, the second experiment completes (the curve goes flat). Thus, the decomposition completes sooner in the presence of manganese(IV) oxide.

The final pressure change should be the same in both experiments. The presence of manganese(IV) oxide does not affect the amount of oxygen formed.



Fig. 37.10 A large amount of oxygen is produced rapidly when manganese(IV) oxide is added to hydrogen peroxide solution

37.6 Reaction rate and effective collisions

For a chemical reaction to take place between two reactant particles, the particles must collide. However, collision alone is not enough for a reaction to occur. The reactant particles have to:

- collide the right way around, i.e. in the right **orientation** (Fig. 37.11), and
- collide with enough energy for bonds to break. The minimum amount of energy required for a chemical reaction to take place is called the **activation energy**.

We will discuss activation energy in Topic 13 Industrial Chemistry.

Reference Website
Refer to the following website for an animation demonstrating the correct and incorrect orientation of reactant particles in a reaction:
http://www.mhhe.com/physsci/chemistry/essentialchemistry/flash/collis11.swf
(accessed August 2014)

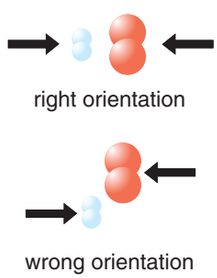


Fig. 37.11 For a reaction to occur, the reactant particles must collide in the right orientation (This is a schematic diagram for illustration only. It does not show the actual collision of the particles.)

orientation 方向　　activation energy 活化能

During a reaction, bonds are broken and new bonds are formed. If a collision is to lead to a reaction, the colliding particles, besides colliding in the right orientation, must also have an amount of kinetic energy equal to or greater than the activation energy to overcome the repulsion between the electron clouds of the reactant particles and to start breaking the bonds (Fig. 37.12). N5

N5 Refer to page T20.

Collisions that lead to a reaction are called **effective collisions**. The rate of a reaction depends on how many effective collisions there are in a unit volume per unit time.

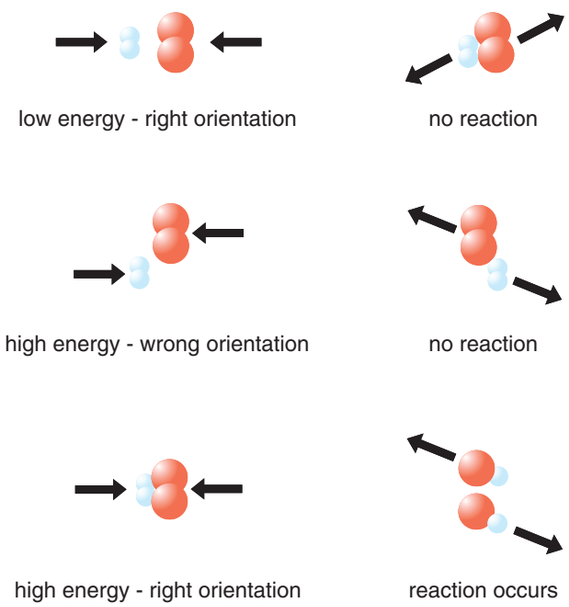


Fig. 37.12 For a reaction to occur, particles must collide in a right orientation and with sufficient kinetic energy

effective collision 有效碰撞

37.7 Why does reaction rate increase with the concentration of reactants?

For many reactions involving solutions or gases, increasing the concentration of reactants increases the rate of the reaction.

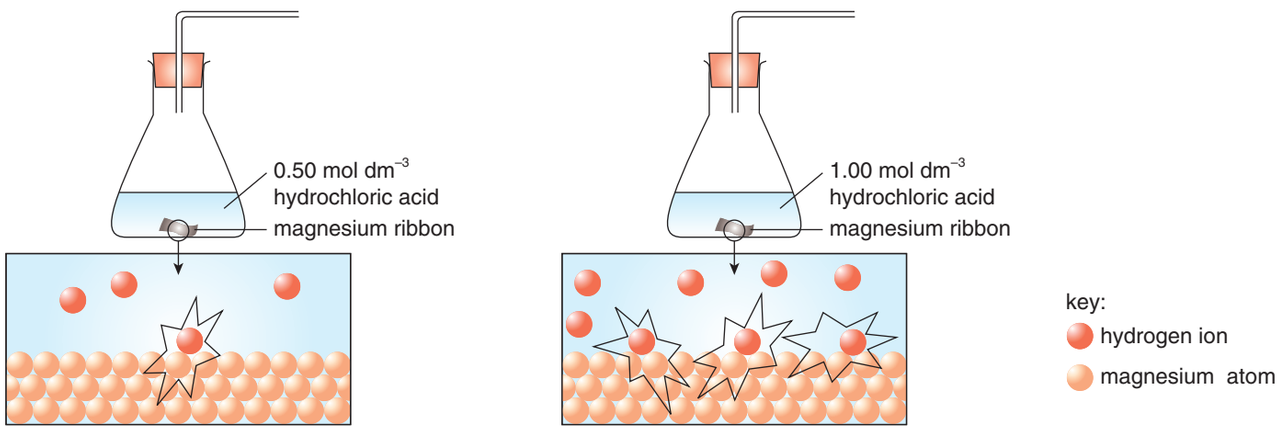
Consider the reaction between magnesium and dilute hydrochloric acid. The reaction involves collisions between magnesium atoms and hydrogen ions.

$$Mg(s) + 2H^+(aq) \longrightarrow Mg^{2+}(aq) + H_2(g)$$

Increasing the concentration of the acid means increasing the number of hydrogen ions per unit volume. The hydrogen ions are more crowded. The chance of collision increases and the number of effective collisions in a unit volume per unit time increases (Fig. 37.13). Hence the rate of the reaction increases.

ASL 2006 Q7(b)

Simulation
Reaction rate and effective collisions — Effect of concentration



(a) Reaction mixture obtained by mixing 0.50 mol dm^{-3} hydrochloric acid and magnesium ribbon

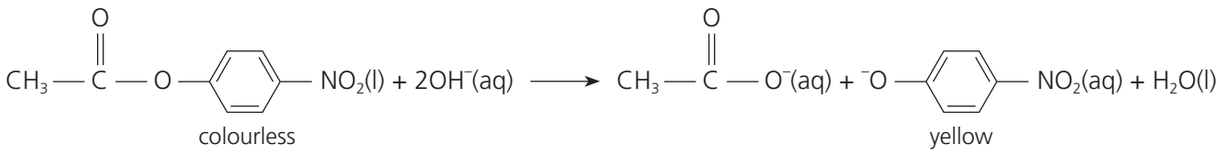
(b) Reaction mixture obtained by mixing 1.00 mol dm^{-3} hydrochloric acid and magnesium ribbon

Fig. 37.13 The greater number of hydrogen ions in (b) leads to more collisions between particles

The idea also helps to explain why the rate of the reaction decreases with time. At the start, there are plenty of hydrogen ions per unit volume. As the hydrogen ions are consumed gradually, there are fewer hydrogen ions per unit volume, i.e. the concentration of the reactant falls. So, the reaction slows down.

Practice 37.4

The ester 4-nitrophenyl ethanoate hydrolyzes in alkaline solution according to the following equation:

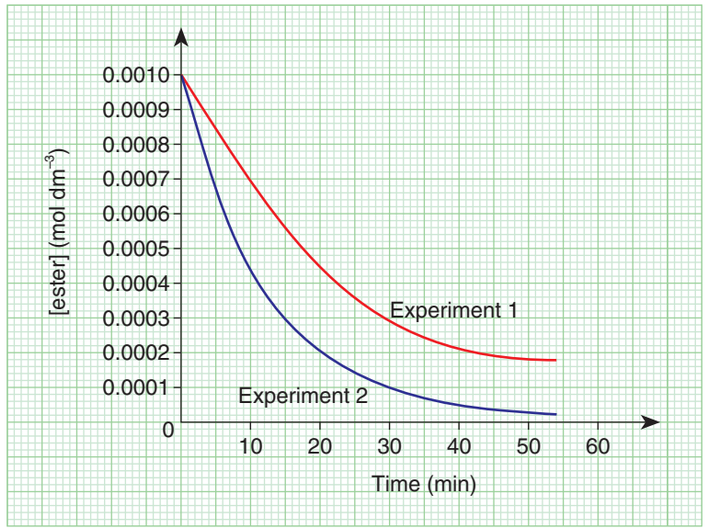


The reaction rate was studied using two solutions of different concentrations of hydroxide ions.

Experiment 1: concentration of $OH^-(aq)$ ions = 0.20 mol dm^{-3}

Experiment 2: concentration of $OH^-(aq)$ ions = 0.40 mol dm^{-3}

The following graphs show how the concentration of the ester, 4-nitrophenyl ethanoate, varied over time in the two experiments.



a) Determine the initial rate of reaction during each experiment.
Experiment 1: 3.3 x 10^{-5} mol dm^{-3} min^{-1}; *Experiment 2*: 6.9 x 10^{-5} mol dm^{-3} min^{-1}; refer to page T21.

b) Describe how the rate of reaction varies with the concentration of $OH^-(aq)$ ions and explain your answer in terms of the behaviour of particles.

Practice 37.4

b) The rate of reaction increases with the concentration of $OH^-(aq)$ ions.
Increasing the concentration of $OH^-(aq)$ ions means increasing the number of reactant particles per unit volume. The particles are more crowded.
The chance of collision increases and the number of effective collisions in a unit volume per unit time increases.

37.8 Why does reaction rate increase with the surface area of a solid reactant?

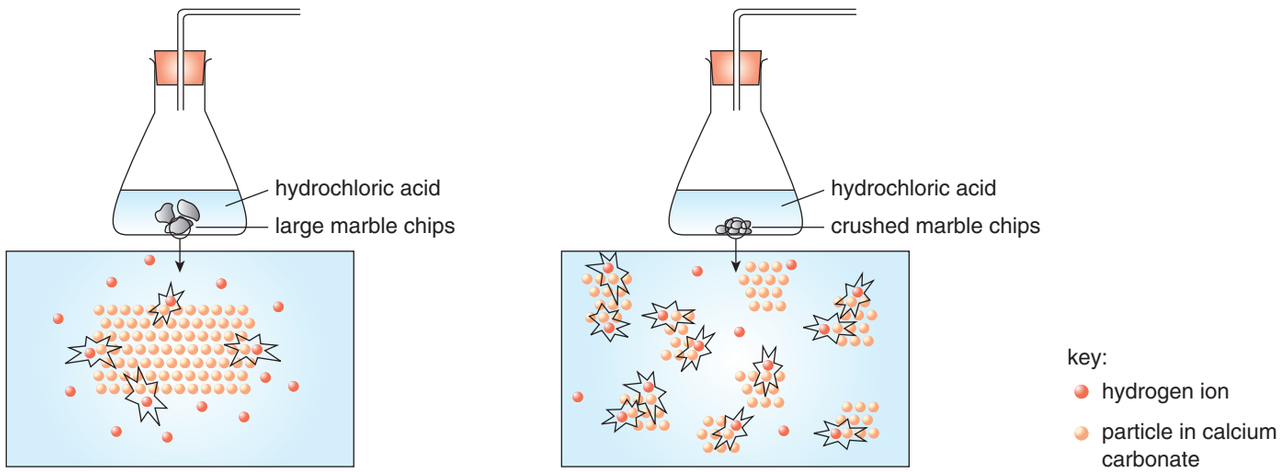
A crushed solid reactant has a greater surface area than a single lump of the same mass. Hence a powdered solid reactant gives a faster reaction than a single lump of the same mass.

ASL 2006 Q7(a)

Consider again the reaction between marble chips and dilute hydrochloric acid.

If large marble chips are used, hydrogen ions from the acid can only hit the particles in the outer layer of the marble chips (Fig. 37.14a). However, if we use crushed marble chips, many more particles are exposed. This increases the contact surface area with the acid. There is a greater chance for collision and the number of effective collisions in a unit volume per unit time increases (Fig. 37.14b). So, the rate of the reaction increases.

Simulation
Reaction rate and effective collisions — Effect of surface area



(a) Hydrogen ions can only hit the particles in calcium carbonate in the outer layer of the large marble chip

(b) When the marble chips are crushed, many more particles in calcium carbonate are exposed and there is a greater chance for collision with hydrogen ions

Fig. 37.14 Increasing the surface area results in a greater chance of collisions

37.9 Why does reaction rate increase with temperature?

In most cases, the rate of a reaction increases when we increase the temperature.

ASL 2005 Q8(a)(iii)
ASL 2006 Q7(c)

If we heat a substance, particles of the substance gain more kinetic energy and move faster. That means the reactant particles collide more often. Besides, the fraction of colliding particles with kinetic energy equal to or greater than the activation energy becomes larger. So, there are more effective collisions in a unit volume per unit time (Fig. 37.15) and the rate of the reaction increases.

Simulation
Reaction rate and effective collisions — Effect of temperature

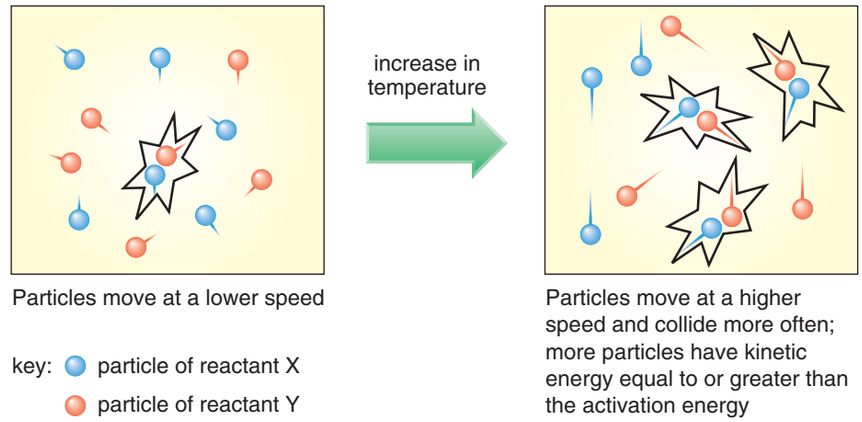


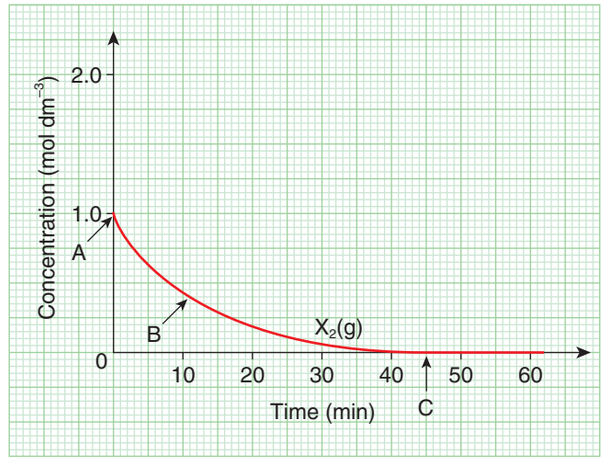
Fig. 37.15 Particles of reactants move faster when the temperature is increased

Example 37.3

Q $X_2(g)$ undergoes decomposition according to the following equation:

$$X_2(g) \longrightarrow 2X(g)$$

In an experiment to study the decomposition of $X_2(g)$, 1.0 mole of $X_2(g)$ was charged into a closed container of volume 1.0 dm^3 kept at a constant temperature. The graph shows the variation of the concentration of $X_2(g)$ in the container with time.



Continued on next page

a) Compare the rates of decomposition of $X_2(g)$ at points A, B and C. Explain why these rates are different.

b) Sketch the variation of the concentration of X(g) with time during the experiment.

CE 2009 Paper 1 Q10(b)(i)

c) Explain why the rate of decomposition of $X_2(g)$ increases at a higher temperature.

A1 a) The rates of decomposition of $X_2(g)$ was A > B > C.

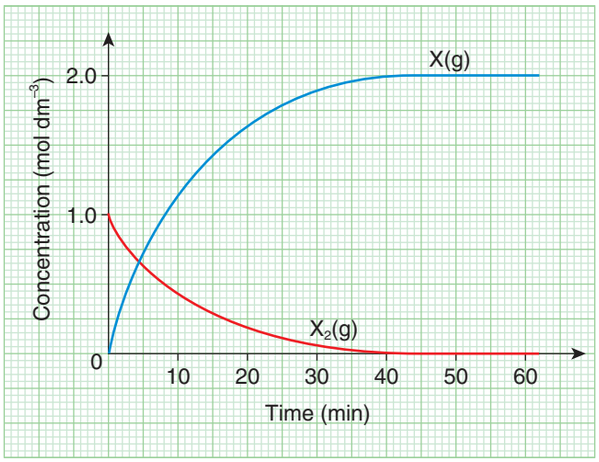
As the reaction proceeded, $X_2(g)$ was consumed gradually. There were fewer particles per unit volume (i.e. the concentration of $X_2(g)$ decreased). So, the decomposition slowed down.

At A, the concentration of $X_2(g)$ was the highest. So, the rate of decomposition was the highest.

At B, the concentration of $X_2(g)$ was lower than that at A. So, the rate of decomposition was lower.

At C, the rate of decomposition was zero as all the $X_2(g)$ was used up.

b)



c) At a higher temperature, the molecules have more kinetic energy and collide more often.

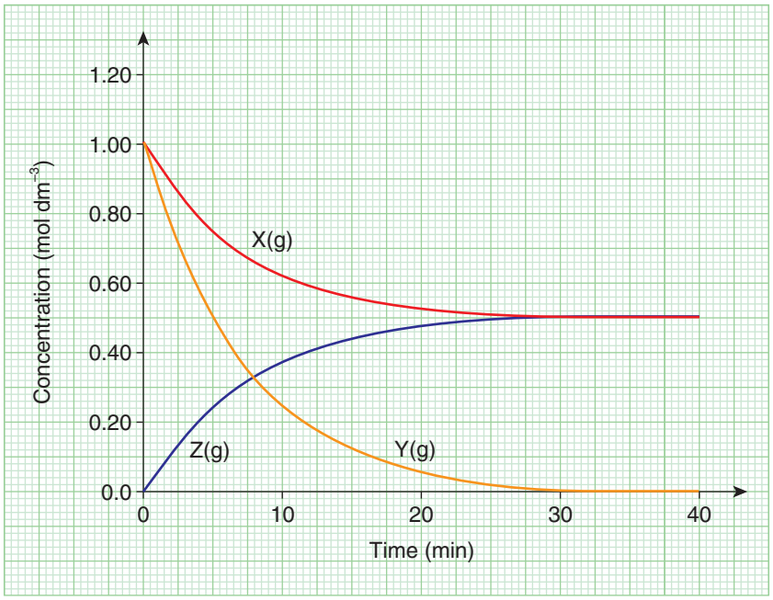
At a higher temperature, the fraction of molecules with kinetic energy equal to or greater than the activation energy becomes larger.

Thus, the rate of decomposition increases.

Practice 37.5

X(g) reacts with Y(g) to give Z(g). A mixture of X(g) and Y(g) was allowed to react in a closed container of volume 1 dm^3 kept at a constant temperature. The graph below shows the changes in concentrations of X(g), Y(g) and Z(g) in the container with time.

(X, Y and Z do not represent symbols of elements.)



a) With reference to the above graph, deduce the chemical equation for the reaction in terms of X(g), Y(g) and Z(g). From the curves, 1 mole of X(g) reacts with 2 moles of Y(g) to give 1 mole of Z(g).
$X(g) + 2Y(g) \longrightarrow Z(g)$

b) Keeping other conditions unchanged, the experiment was repeated by allowing the mixture of X(g) and Y(g) to react in a container of volume 2 dm^3 instead. Would the time required for the reaction to complete remain the same? Explain your answer in terms of collisions between particles of reactants.
The time would be longer.
In a larger container, the concentrations of the reactants become smaller. The chance of collision decreases and hence the rate of the reaction decreases.

Discussion
The rate of a reaction depends on how many effective collisions between reactant particles there are in a unit volume per unit time.
Only those colliding particles with an energy equal to or greater than the activation energy can react.
When the temperature increases, the reactant particles have more kinetic energy and collide more often. More importantly, a larger portion of the particles have an energy equal to or greater than the activation energy.

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

A student made the following remark:

'The rate of reaction increases with temperature because the kinetic energy of the reactant particles increases with temperature.'

Is the explanation provided by the student regarding the increase in reaction rate appropriate? Discuss.

37.10 Applications of catalysts

We will further discuss catalysts in Topic 12 Patterns in the Chemical World and Topic 13 Industrial Chemistry.

We will discuss the Haber process in Topic 11 Chemical Equilibrium and Topic 13 Industrial Chemistry.

We have discussed catalytic converters in Topic 7 Fossil Fuels and Carbon Compounds.

Catalysts are widely used in industries (Fig. 37.16). The main reason for using catalysts is to increase the rate of a reaction. Increasing the rate of a reaction shortens the production time and hence increases the production efficiency (Table 37.1).

A reaction may be carried out at a lower temperature and pressure in the presence of a catalyst. As a result, the amount of energy needed to fuel the reaction decreases and costs become lower.

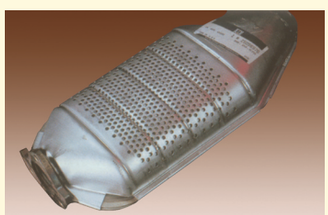


Fig. 37.16 Catalysts are made in all kinds of sizes and shapes

N6 Refer to page T21.

Table 37.1

Examples of industrial catalyst N6

Process	Catalyst(s) used	Reaction(s) involved	Use
Haber process	iron	$N_2(g) + 3H_2(g) \overset{Fe}{\rightleftharpoons} 2NH_3(g)$	manufacture of ammonia
Contact process	platinum or vanadium(V) oxide	$2SO_2(g) + O_2(g) \overset{Pt \text{ or } V_2O_5}{\rightleftharpoons} 2SO_3(g)$	manufacture of sulphuric acid
Reactions in catalytic converters  **Fig. 37.17** A catalytic converter	platinum (or palladium) and rhodium	$2CO(g) + 2NO(g) \xrightarrow{Rh} 2CO_2(g) + N_2(g)$ $2CO(g) + O_2(g) \xrightarrow{Pt} 2CO_2(g)$ $2C_8H_{18}(l) + 25O_2(g) \xrightarrow{Pt} 16CO_2(g) + 18H_2O(l)$ unburnt hydrocarbons (e.g. octane)	converting harmful pollutants in car exhaust gas to relatively harmless nitrogen, carbon dioxide and water

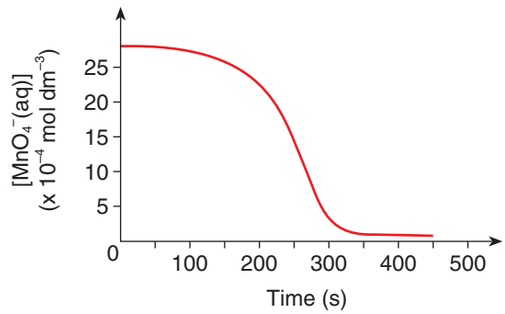
Haber process 哈柏法　　Contact process 接觸法

Example 37.4

Q Potassium permanganate reacts with sodium ethanedioate under acidic conditions according to the following equation:

$$2MnO_4^-(aq) + 5C_2O_4^{2-}(aq) + 16H^+(aq) \longrightarrow 2Mn^{2+}(aq) + 10CO_2(g) + 8H_2O(l)$$

A student followed the progress of this reaction using a colorimeter. The results obtained are shown in the graph below.



Based on the experimental results, the student suggested that one of the products, $Mn^{2+}(aq)$, is a catalyst for the reaction.

a) What evidence from the above graph supports the student's suggestion? AL 2010 Paper 2 Q2(b)(ii)(II)

b) Suggest how the student can show whether or not $Mn^{2+}(aq)$ is a catalyst for the reaction.

A a) The rate of consumption of $MnO_4^-(aq)$ ions is low at the start and then increases rapidly.

It is likely to be due to the building up of the concentration of $Mn^{2+}(aq)$ which catalyzes the reaction.

b) Repeat the experiment with a few drops of $Mn^{2+}(aq)$ added to the reaction mixture.

The consumption of $MnO_4^-(aq)$ ions will be faster at the start if $Mn^{2+}(aq)$ is a catalyst.

Do you know

Shrimp shells help make biodiesel

As concerns over global warming and dwindling fossil fuel reserves increase, more and more interest is growing in renewable fuels such as biodiesel to meet energy demands. However, today's biodiesel production processes require catalysts to speed up the chemical reactions that convert plant oils into diesel fuel, and traditional catalysts cannot be reused.

Scientists in China have reported the development of a catalyst made from shrimp shells that can convert canola oil to biodiesel faster and more efficiently than some conventional catalysts do. The new catalyst can be reused and the manufacturing process minimizes waste production and pollution.

Do you know

Biological washing powders

Biological washing powders contain enzymes that help break down proteins, fat and starch. This action helps remove stains such as chocolate or hamburger grease.

The enzymes in biological washing powders enable effective cleansing at lower temperatures. This reduces the energy costs and also causes less damage to the clothes being washed.

Stains are made of different types of molecules and a range of enzymes are needed to break them down. For example, proteases break down proteins and remove stains caused by blood, egg and other proteins. Lipases break down fats and grease.

37.11 Enzymes

Enzymes are protein-based catalysts found in living things. The biological processes that occur within all living things are chemical reactions, and most of them are regulated by enzymes.

Enzymes are extremely effective catalysts. They are also extremely specific: one enzyme normally catalyzes one reaction of a specific molecule.

An enzyme works best between 25 °C and 45 °C. At higher temperatures, it stops working and becomes **denatured**. At lower temperatures, it becomes inactive.

A huge industry has grown up around the use of enzymes. For example, biological washing powders contain enzymes to digest protein-based stains (Fig. 37.18). Baby food also contains enzymes to 'pre-digest' the protein in the food (Fig. 37.19).



Fig. 37.18 These biological washing powders contain enzymes



Fig. 37.19 Baby foods contain enzymes

Many industrial processes, including the manufacture of beer, yogurt, cheese and medicines use enzymes as well (Fig. 37.20).



Fig. 37.20 Enzymes are used in the manufacture of cheese

enzyme 酶　　denatured 變性

Yeast and fermentation

We will further discuss fermentation in Topic 13 Industrial Chemistry.

Human beings have used enzymes for thousands of years. For example, enzymes produced by **yeast** (Fig. 37.21) are used in the brewing industry. Enzymes from yeast catalyze the conversion of sugars to ethanol. The process is called **fermentation**.

$$\underset{\text{sugar (glucose)}}{C_6H_{12}O_6(aq)} \xrightarrow{\text{enzymes in yeast}} \underset{\text{ethanol}}{2C_2H_5OH(aq)} + \underset{\text{carbon dioxide}}{2CO_2(g)}$$

✓ Fermentation is the slow breakdown of large organic molecules into small molecules by microorganisms in the absence of oxygen.

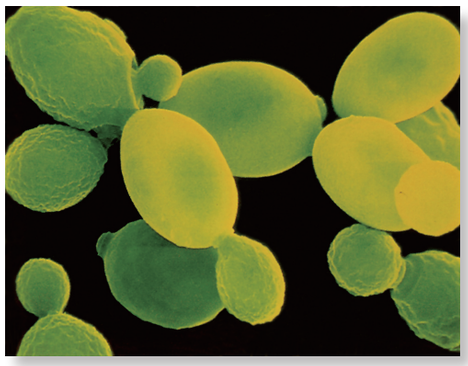


Fig. 37.21 Yeast cells

yeast 酵母　　fermentation 發酵作用

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

Chemistry Magazine

Saved by a very fast chemical reaction

Airbags are a familiar addition to car safety these days. One type of airbag uses a very fast gas-producing chemical reaction to blow up the bag (Fig. 37.22).



Fig. 37.22 Airbags are blown up by a very fast gas-producing reaction

How does an airbag work?

The airbag is made up of three parts. Firstly, there is the bag itself. The second part is the crash sensor. Finally, there is the inflation system.

When a car decelerates very quickly, as in a head-on crash, the crash sensor would send an electric spark to the inflator. This activates the decomposition of the sodium azide (NaN_3) sealed inside the inflator. The nitrogen gas produced inflates the airbag, cushioning the driver and reducing the likelihood of injury.

The process, from the initial impact of the crash to full inflation of the airbag, takes only about 40 milliseconds.

Questions

1 When there is a crash, sodium azide (NaN_3) inside the inflator decomposes into its constituent elements.

Write the chemical equation for the decomposition of sodium azide. ASL 2011 Paper 2 Q2(c) AL 2011 Paper 2 Q2(c)

2 Suggest why a driver in a car may suffer skin burns if the inflated airbags burst.

3 Airbags have been in the news in recent years because of deaths resulting from airbag deployments in relatively minor low speed crashes.

Discuss how airbag deployments may lead to death.

1 $2NaN_3(s) \longrightarrow 2Na(s) + 3N_2(g)$

2 Any one of the following:
- The Na(s) produced will react with water vapour in the air / water from the skin to give corrosive NaOH(aq) which causes skin burns.
- The reaction between Na(s) and water is highly exothermic.

3 Airbags had been designed based on car occupants of average size, so the force and direction of impact of the bag itself could hurt children or small adults.
Car manuals say young children should not be put in front seats where they might be injured or killed by an inflating airbag.

Have you mastered?

Key terms

surface area 表面面積 34　　crushed marble chips 大理石碎粒 34　　catalyst 催化劑 39

orientation 方向 41　　activation energy 活化能 41　　effective collision 有效碰撞 42

Haber process 哈柏法 49　　Contact process 接觸法 49　　enzyme 酶 51　　denatured 變性 51

yeast 酵母 52　　fermentation 發酵作用 52

Checklist

After studying this unit, you should be able to

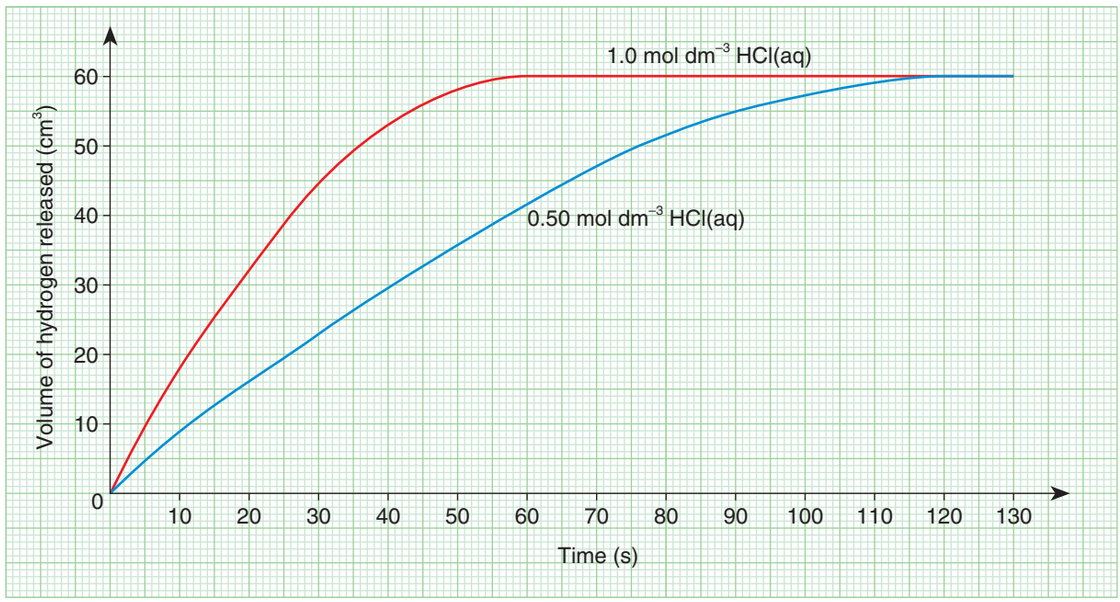
- ☐ design and perform experiments to study the effects of concentration, surface area, temperature and catalyst on rate of reaction;
- ☐ interpret results (e.g. graphs) collected through investigations on factors affecting rate of reaction;
- ☐ explain the effect of changes in concentration, surface area and temperature on rate of reaction with reference to the collision theory;
- ☐ appreciate the importance of catalysts in chemical industries and biological systems.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 In most cases, the rate of a reaction increases when the concentration of a reactant is increased.

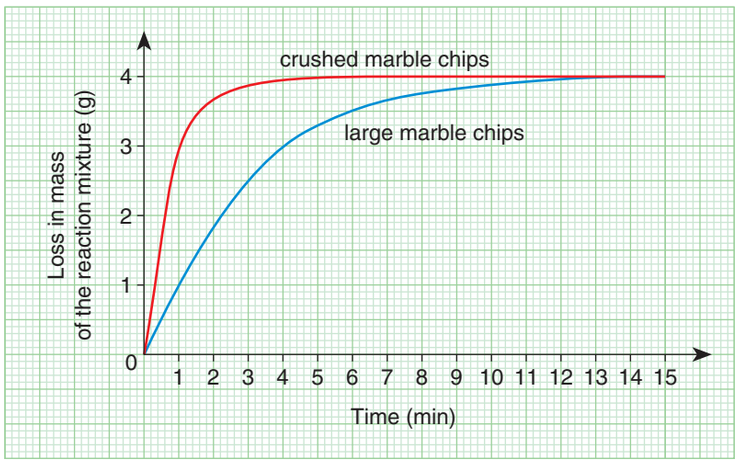
The following curves show the volume of hydrogen released in the reactions between magnesium ribbons of equal length and excess hydrochloric acid of different concentrations.



2 The rate of a reaction increases when the surface area of a solid reactant is increased.

A powdered solid reactant has a greater surface area than a single lump of the same mass.

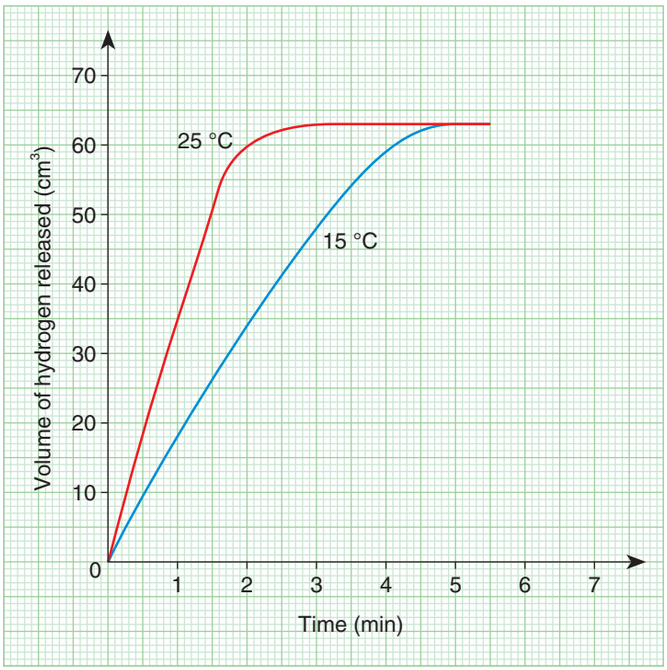
The following curves show the loss in mass of the reaction mixture in the reactions between equal masses of marble chips of different surface areas and excess dilute hydrochloric acid.



Have you mastered?

3 In most cases, the rate of a reaction increases when the temperature is increased.

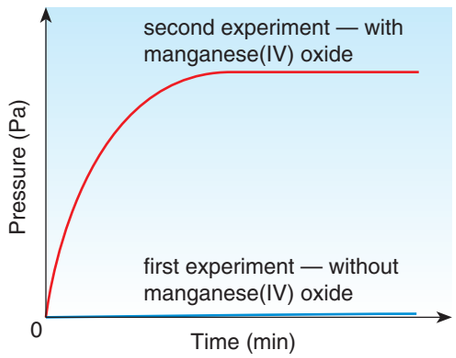
The following curves show the volume of hydrogen released in the reactions between magnesium ribbons of equal length and excess dilute hydrochloric acid at different temperatures.



Have you mastered?

4 A catalyst is a substance which alters the rate of a reaction without itself undergoing any permanent chemical changes.

The following curves show the pressure change during the decomposition of hydrogen peroxide with / without the catalyst manganese(IV) oxide.



5 The rate of a reaction depends on how many effective collisions there are in a unit volume per unit time. To determine whether a collision is effective, two factors must be considered:

- whether the reactant particles collide in the right orientation;
- whether the colliding particles have kinetic energy equal to or greater than the activation energy.

6 Increasing the concentration of a reactant means increasing the number of reactant particles per unit volume. The chance of collision increases and the number of effective collisions in a unit volume per unit time increases. Hence the rate of the reaction increases.

7 A powdered solid reactant has a greater surface area than a single lump of the same mass.

Increasing the surface area of a solid reactant increases the contact surface area between the reactant particles. There is a greater chance for collision and the number of effective collisions in a unit volume per unit time increases. Hence the rate of the reaction increases.

8 When the temperature is increased, the reactant particles have more energy and collide more often. Besides, the fraction of colliding particles with kinetic energy equal to or greater than the activation energy becomes larger. So, there are more effective collisions in a unit volume per unit time and the rate of the reaction increases.

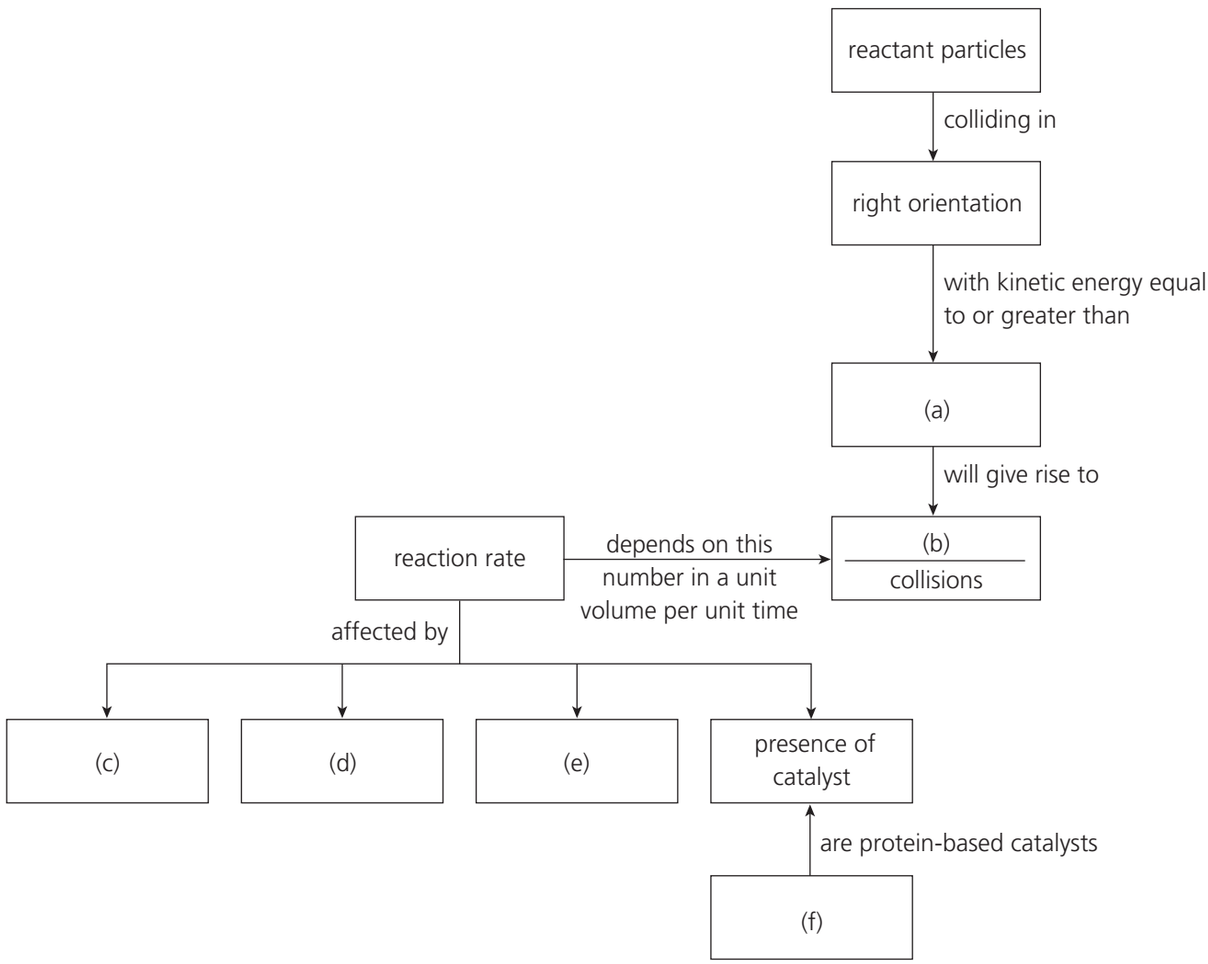
9 Enzymes are protein-based catalysts found in living things.

Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.



a) activation energy
b) effective
c) concentration of reactant
d) surface area of solid reactant
e) temperature
f) enzymes

2 Consider the reaction between calcium carbonate lumps and hydrochloric acid. How would the rate of carbon dioxide production be affected by each of the following changes?

	Rate increases / remains the same / decreases
Using powdered calcium carbonate	Rate increases
Using an acid of higher concentration	Rate increases
Adding water to the reaction mixture	Rate decreases
Adding more acid of the same concentration	Rate remains the same
Raising the temperature of the acid	Rate increases

3 Explain the following statements in terms of reaction rates:

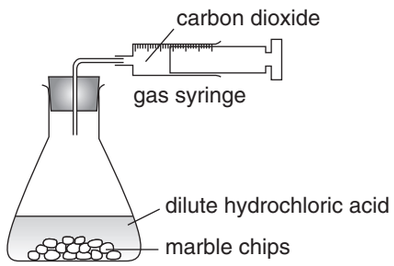
a) Potato chips cook faster in oil than whole potatoes do.
Cutting whole potatoes into chips increases the surface area of potatoes, thus increases the cooking rate.

b) Food products are often stored in refrigerators.
Low temperatures slow down reactions that lead to the spoiling of food. Thus, storing food products in refrigerators helps keep food from spoiling.

Part II Multiple choice questions

Directions: Questions 4 and 5 refer to the following experiment.

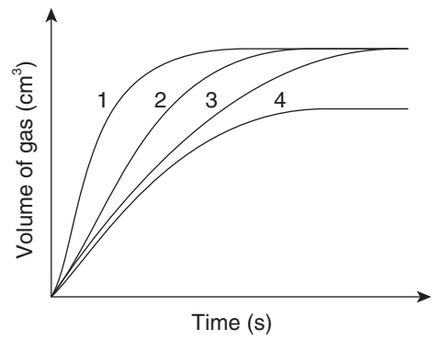
A student investigated the rate of reaction between dilute hydrochloric acid and marble chips (calcium carbonate) using the apparatus shown.



Four separate *experiments*, *Experiments 1*, *2*, *3* and *4*, were carried out.

In each experiment the volume of carbon dioxide in the gas syringe was measured every 30 seconds. The conditions in all four experiments were the same except that different sized marble chips were used.

The results of the experiments are shown on the graph.



4 In which experiment were the smallest marble chips used?

(A) *Experiment 1*
B *Experiment 2*
C *Experiment 3*
D *Experiment 4*

The reaction rate in *Experiment 1* was the highest. Thus, the smallest marble chips were used.

5 In which experiment did some carbon dioxide leak from the gas syringe during the experiment?

A *Experiment 1*
B *Experiment 2*
C *Experiment 3*
(D) *Experiment 4*

The volume of gas in *Experiment 4* was lower than the other three. Thus, some carbon dioxide leaked from the gas syringe.

(Edexcel GCSE (Foundation and Higher Tiers), Chemistry, Unit C2, Jun. 2009, 21–22)

Unit Exercise

6 Ammonium nitrite decomposes when heated according to the equation:

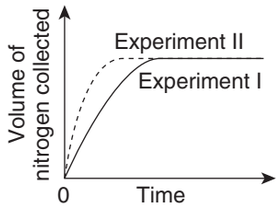
$NH_4NO_2(aq) \longrightarrow 2H_2O(l) + N_2(g)$

Two samples of 100 cm^3 of ammonium nitrite solution of different concentrations were allowed to undergo decomposition.

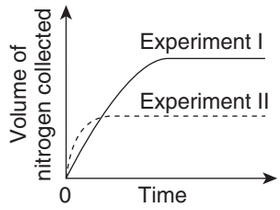
Experiment	Concentration of ammonium nitrite solution
I	5.0%
II	2.5%

The volume of nitrogen collected was measured at regular time intervals in each case. Which of the following graphs best represents the experimental results?

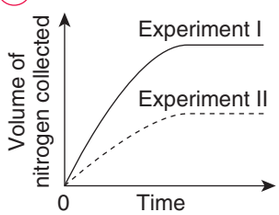
A



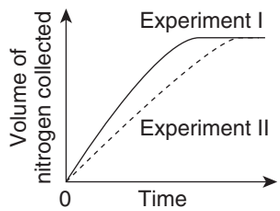
B



C



D



7 The rates of chemical reactions are affected by the average energy and the frequency of the collisions between the particles.

Which row correctly describes the effect on these factors of using a more concentrated hydrochloric acid?

	Effect on average energy of collisions	Effect on frequency of collisions
A	increased	increased
B	increased	no change
C	no change	increased
D	no change	no change

(Edexcel GCSE (Higher Tier), Chemistry, Unit C2, Nov. 2011, 36)

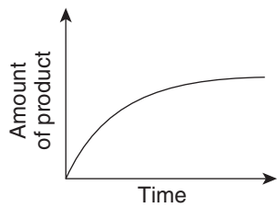
8 Which row shows the effect of a lower temperature on the frequency and energy of particle collisions during a chemical reaction?

	Frequency of particle collisions	Energy of particle collisions
A	decreased at lower temperature	decreased at lower temperature
B	unchanged at lower temperature	decreased at lower temperature
C	unchanged at lower temperature	unchanged at lower temperature
D	decreased at lower temperature	unchanged at lower temperature

(Edexcel GCSE (Higher Tier), Chemistry, Unit C2, Mar. 2010, 28)

When the temperature is lowered, both the frequency and energy of particle collisions decrease.

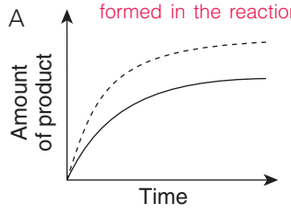
9 The curve shown below shows the results obtained in a chemical reaction.



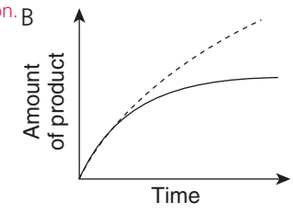
In a second experiment, a catalyst was added to the reaction mixture. All other conditions were kept the same. Which of the following graphs shows the experimental results (dotted line) obtained using the catalyst?

A catalyst can increase the rate of the reaction.
A catalyst does not change the amount of product formed in the reaction.

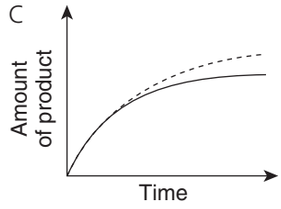
A



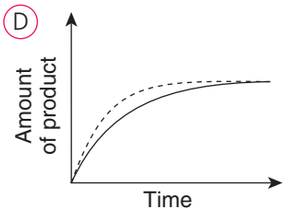
B



C



D



6 The rate of decomposition in *Experiment I* was higher because a higher concentration of ammonium nitrite solution was used.
The volume of nitrogen collected in *Experiment I* was higher because a greater amount of ammonium nitrite was used.

7 Using a more concentrated hydrochloric acid increases the frequency of collisions.
However, the average energy of collisions does NOT change because the temperature remains the same.

10 In an experiment, 10 g of zinc granules are added to 100 cm^3 of 1 M HCl(aq) in a beaker. Which of the following changes to the experiment can increase the initial rate of the reaction?

(1) Use 200 cm^3 of 1 M HCl(aq) to replace 100 cm^3 of 1 M HCl(aq).
(2) Use 50 cm^3 of 2 M HCl(aq) to replace 100 cm^3 of 1 M HCl(aq).
(3) 10 g of zinc granules of greater size are used instead.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2010, 44)

11 Some brands of washing powder contain enzymes. Which of the following statements about the action of the enzymes is / are correct?

(1) The activity of the enzymes increases with temperature.
(2) The enzymes facilitate the removal of specific kinds of dirt.
(3) The enzymes reduce the surface tension of water.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKDSE, Practice paper 1A, 2012, 32)

Part III Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

12 Richard was investigating the reaction between magnesium and dilute hydrochloric acid.

He used magnesium ribbon and magnesium powder in his investigation.

Richard measured the temperature of the dilute hydrochloric acid and then added 0.5 g of magnesium ribbon.

After one minute, he measured the temperature again. An exothermic reaction had taken place.

He repeated the experiment using 0.5 g of magnesium powder instead of the ribbon.

Here are his results.

	Temperature of the acid before adding the magnesium (°C)	**Temperature of the reaction mixture after one minute (°C)**
Magnesium ribbon	18	23
Magnesium powder	18	49

a) How do these results show the reaction is exothermic?

b) How could Richard check that his results are reliable?

c) Richard found the reaction with the magnesium powder was faster than with magnesium ribbon.

Why does magnesium powder react faster?

d) Suggest a way for Richard to make the reaction between 0.5 g magnesium ribbon and dilute hydrochloric acid slower.

(Edexcel GCSE (Foundation Tier), Chemistry, Unit C2, Mar. 2009, 4)

12 a) The temperature rose.
b) Repeat the experiment to obtain similar results.
c) The contact surface area between magnesium and the acid increases.
There is a greater chance for collision and the number of effective collisions in a unit volume per unit time increases.
d) Dilute the acid / cooler temperature of acid / fold magnesium ribbon

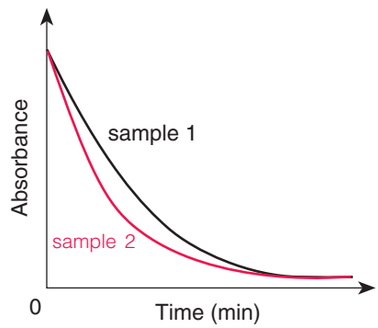
13 Acidified propanone solution reacts with iodine according to the following equation:

$$CH_3COCH_3(aq) + I_2(aq) \longrightarrow CH_3COCH_2I(aq) + H^+(aq) + I^-(aq)$$

In an experiment to determine the effect of change in concentration of propanone on the rate of the reaction, a student prepared two samples as listed below:

Sample	1	2
Volume of 1 mol dm^{-3} propanone (cm^3)	0.75	1.50
Volume of 1 mol dm^{-3} sulphuric acid (cm^3)	0.75	0.75
Volume of distilled water (cm^3)	1.50	0.75
Number of drops of 0.02 mol dm^{-3} I_2	30	30

The student used a colorimeter to follow the progress of the reaction. The results obtained for sample 1 are shown in the following graph:



a) Explain why the volume of each sample was kept constant. The concentration of propanone in each sample is directly proportional to the volume of propanone solution used.

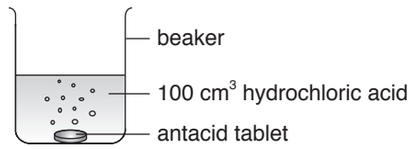
b) Account for the change of absorbance for sample 1 as shown in the graph.

c) Sketch, on the same graph, the expected result for sample 2. When the reaction proceeds, the brown colour of the reaction mixture becomes less intense as the concentration of iodine falls.

14 Emily investigates antacid tablets. Antacids neutralize excess acid in your stomach.

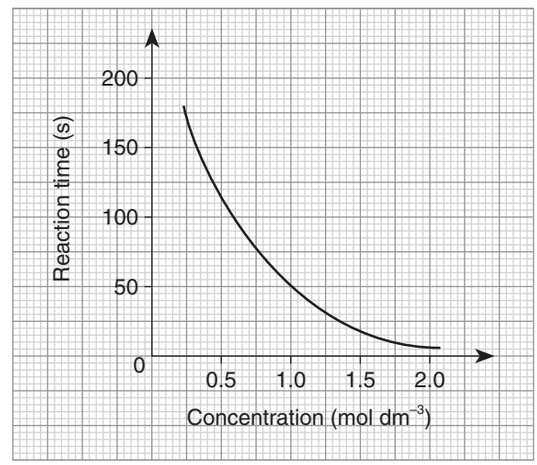
Emily uses one tablet in each experiment. She adds the tablet to 100 cm^3 of hydrochloric acid and measures the time it takes to fully react. This is the reaction time.

Look at the diagram. It shows the apparatus she uses.



Emily does the experiment several times. Each time she uses a different concentration of acid.

Look at the graph of Emily's results.



a) Emily uses hydrochloric acid with a concentration of 1.5 mol dm^{-3}.

Look at the graph above. What is the reaction time? 18 seconds

b) Emily finds that the reaction time is shorter with concentrated acid than with dilute acid.

Explain why. Use ideas about particles.

c) Emily repeats the experiment using a crushed tablet. She finds that the reaction is faster.

Explain why. Use ideas about collisions between particles.

14 b) The number of acid particles per unit volume increases. There is a greater chance for collision and the number of effective collisions in a unit volume per unit time increases.

c) The contact surface area between the tablet and hydrochloric acid increases.
There is a greater chance for collision and the number of effective collisions in a unit volume per unit time increases.

Unit Exercise

d) These antacid tablets contain calcium carbonate.

In this reaction, calcium carbonate, $CaCO_3$, reacts with hydrochloric acid, HCl. Calcium chloride, $CaCl_2$, carbon dioxide and water are made.

Write a balanced chemical equation for this reaction. $CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 1, Jan. 2011, 8)

15 When dilute hydrochloric acid is added to sodium thiosulphate solution, the following reaction takes place. Refer to page T22.

$$Na_2S_2O_3(aq) + 2HCl(aq) \longrightarrow 2NaCl(aq) + SO_2(g) + H_2O(l) + S(s)$$

An experiment was carried out to study the effect of varying the concentration of sodium thiosulphate solution on the rate of reaction. Some of the volumes of solutions used are shown below.

Volume of 0.1 mol dm^{-3} $Na_2S_2O_3(aq)$ (cm^3)	Volume of water (cm^3)	Volume of 0.1 mol dm^{-3} HCl(aq) (cm^3)	Reaction time (s)
100	0	5	20
80	20	5	25
60	40	5	33
40	60	5	50
20	80	5	100

a) Complete the table to show the volumes of water and acid that would have been used.

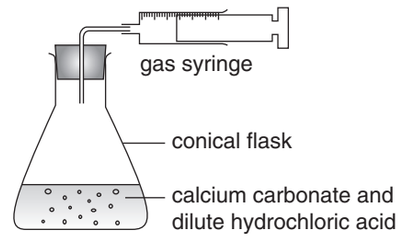
b) Describe how the reaction time could be measured. Draw a diagram of the experimental set-up.

c) Describe how the relative rate of reaction could be obtained from each result.

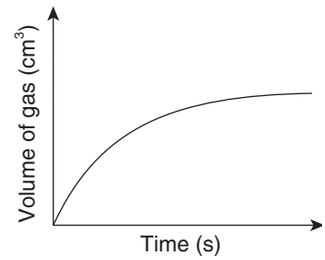
d) Describe how the rate of reaction varies with the concentration of sodium thiosulphate solution and explain your answer in terms of the behaviour of particles.

16 a) The rate of the reaction is the highest at the start.
The rate of the reaction decreases as the reaction proceeds and the reaction stops finally.
b) The change: acid of lower concentration is used.
Explanation: the rate of reaction is lower;
less product is formed;
it takes shorter time for the reaction to end.

16 Jake carries out an experiment to investigate the rate of reaction between calcium carbonate and dilute hydrochloric acid.



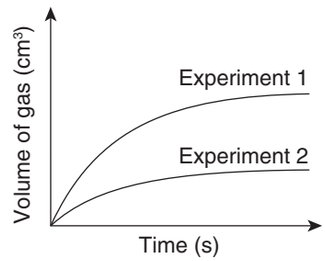
This is a graph of Jake's results.



a) Describe what happens to the rate of the reaction during the experiment.

b) Jake does a second experiment, *Experiment 2*.

Jake plots the results of his second experiment.



These are the conditions that Jake used for his first experiment.

Conditions for his first experiment
acid volume: 25 cm^3
acid concentration: 10 g dm^{-3}
calcium carbonate: 5.0 g, large pieces

Jake used the same mass of calcium carbonate in *Experiment 2*.

Suggest and explain one change that Jake makes to the conditions for *Experiment 2*.

(OCR GCSE 21st Century Science (Higher Tier), Chem. A, Unit 2, Jun. 2011, 4(a), (b))

17 Monty investigates the properties of two acids:

- dilute ethanoic acid, CH_3COOH; and
- dilute hydrochloric acid, HCl.

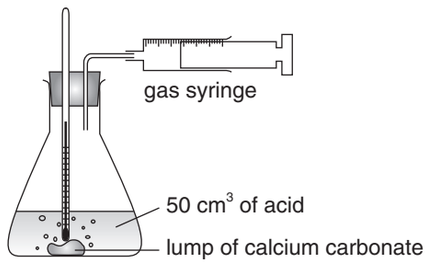
a) Monty adds a small piece of magnesium ribbon to dilute ethanoic acid.

Monty sees bubbles of a gas. At the end of the reaction, a colourless solution is left. The colourless solution contains magnesium ethanoate, $Mg(CH_3COO)_2$.

Write down the balanced chemical equation for the reaction between magnesium and ethanoic acid.
$Mg(s) + 2CH_3COOH(aq) \longrightarrow Mg(CH_3COO)_2(aq) + H_2(g)$

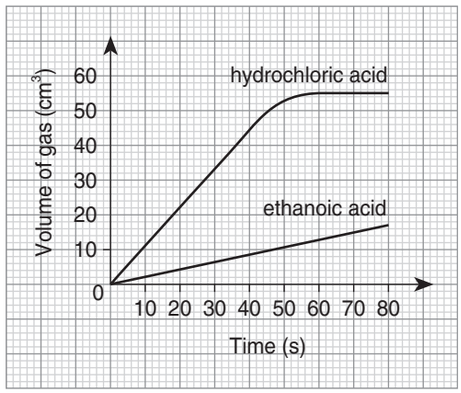
b) Monty investigates the reaction of both acids with a lump of calcium carbonate. He wants to find out the volume of gas made every 10 seconds.

Look at the apparatus he uses.



He does two experiments, one with dilute ethanoic acid and one with dilute hydrochloric acid. He makes sure he does a fair test.

Look at the graph of his results.



i) The reaction between calcium carbonate and ethanoic acid is still happening after 80 seconds.

What will be the total volume of gas collected at the end of this reaction? 55 cm^3

ii) Dilute hydrochloric acid reacts much faster than dilute ethanoic acid. Explain why.
Refer to page T22.
Use ideas about

- hydrogen ions;
- collisions between particles.

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 2, Jan. 2008, 6)

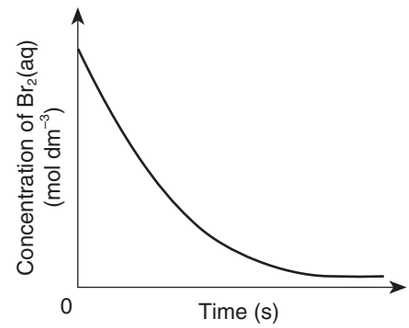
18 Consider the reaction below:
Refer to page T23.

$Br_2(aq) + HCOOH(aq) \longrightarrow 2Br^-(aq) + 2H^+(aq) + CO_2(g)$

The table below lists the experimental data obtained at a certain temperature:

Sample	Volume used (cm^3)			Relative initial rate for the disappearance of $Br_2(aq)$
	0.010 mol dm^{-3} $Br_2(aq)$	0.20 mol dm^{-3} HCOOH(aq)	$H_2O(l)$	
1	2.0	10.0	8.0	1
2	4.0	10.0	6.0	2

a) The concentration of $Br_2(aq)$ in sample 1 is determined at different times. The graph below shows the results obtained.



Suggest how the initial rate of disappearance of $Br_2(aq)$ can be determined from the graph above.

b) Why is it necessary to keep the concentration of HCOOH(aq) much higher than that of $Br_2(aq)$?

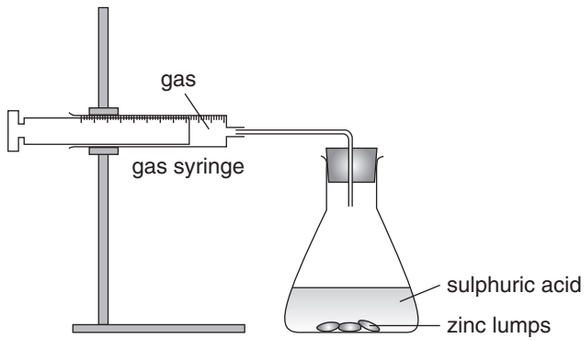
c) Describe how the rate of reaction varies with the concentration of bromine and explain your answer in terms of the behaviour of particles.

Unit Exercise

19 Paul investigates the reaction between sulphuric acid and zinc metal.

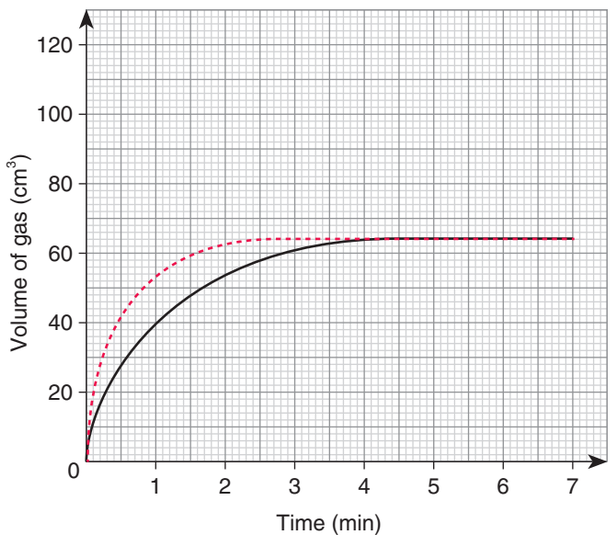
Hydrogen gas and zinc sulphate are made.

a) Look at the diagram. It shows the apparatus he uses.



He measures the volume of gas in the gas syringe every minute.

Look at the graph. It shows his results.



i) At what time did the reaction finish?

4.5 minutes

ii) Paul uses a catalyst to speed up the reaction.

He does not change anything else.

Sketch what his new results may look like.

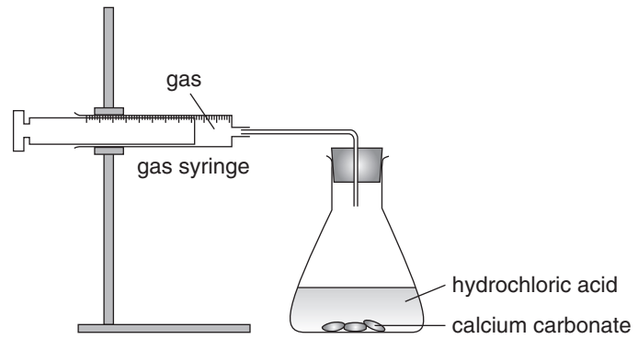
b) This reaction can also be speeded up by using zinc powder instead of zinc lumps. Explain why.

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 1, Jan. 2009, 6(b)–(c))

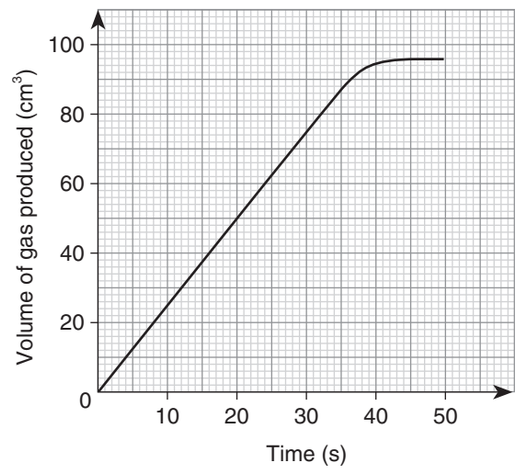
19 b) The contact surface area between zinc and sulphuric acid increases.
There is a greater chance for collision and the number of effective collisions in a unit volume per unit time increases.

20 John reacts calcium carbonate with hydrochloric acid. Carbon dioxide gas is made and collected.

The diagram shows the apparatus he uses.



a) Look at the graph. It shows his results.



Calculate the average rate of reaction during the first 30 seconds.

b) The reaction goes faster at higher temperatures. Explain why.

Use ideas about collisions between particles.

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 1, Jun. 2009, 9)

20 a) Rate of reaction during the first 30 seconds $= \frac{74\ cm^3}{30\ s}$
$= 2.5\ cm^3\ s^{-1}$

b) The rate of a reaction depends on how many effective collisions between reactant particles there are in a unit volume per unit time.
Only those colliding particles with an energy equal to or greater than the activation energy can react.
At a higher temperature, the reactant particles have more kinetic energy and collide more often.
More importantly, a larger portion of the particles have an energy equal to or greater than the activation energy.

Unit Exercise

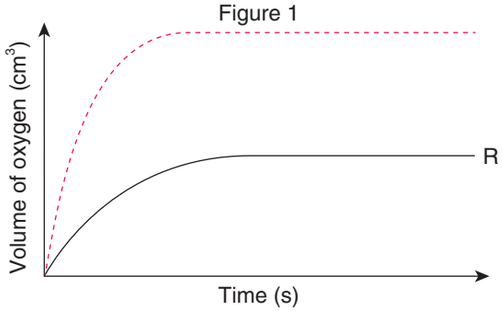
21 A student carried out an experiment to determine the rate of decomposition of hydrogen peroxide into water and oxygen gas.

The student used 100 cm^3 of a 1.0 mol dm^{-3} solution of hydrogen peroxide at 298 K and measured the volume of oxygen collected.

Curve R, in each of Figures 1, 2 and 3, shows how the volume of oxygen collected changed with time under these conditions.

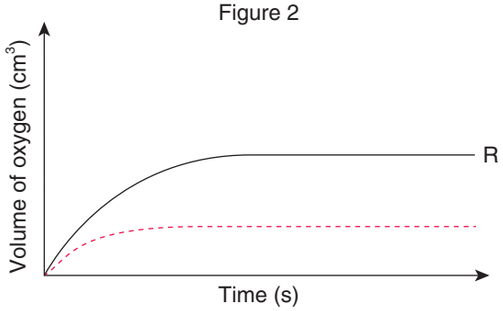
a) Draw a curve on Figure 1 to show how the volume of oxygen collected will change with time if the experiment is repeated at 298 K using 100 cm^3 of a 2.0 mol dm^{-3} solution of hydrogen peroxide.

Figure 1



b) Draw a curve on Figure 2 to show how the volume of oxygen collected will change with time if the experiment is repeated at 298 K using 100 cm^3 of a 0.4 mol dm^{-3} solution of hydrogen peroxide.

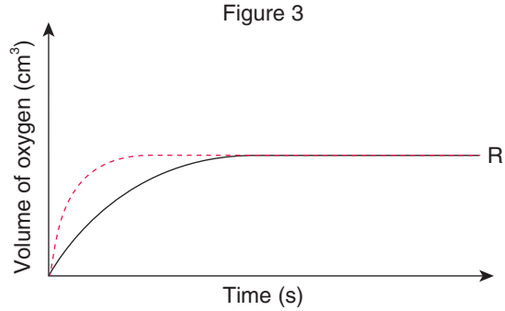
Figure 2



c) Draw a curve on Figure 3 to show how the volume of oxygen collected will change with time if the original experiment is repeated at a temperature higher than 298 K.

You should assume that the gas is collected at a temperature of 298 K.

Figure 3



d) Explain why the slope of curve R decreases as time increases.

e) The student discovered that hydrogen peroxide decomposes at a faster rate when a few drops of hydrogen bromide solution are added to the solution.

He found on the internet that this decomposition is thought to proceed in two steps as shown by the following equations:

Step 1 $H_2O_2 + HBr \longrightarrow HBrO + H_2O$

Step 2 $HBrO + H_2O_2 \longrightarrow H_2O + O_2 + HBr$

i) Write an equation for the overall reaction.

ii) Give ONE reason, other than the increase in rate of reaction, why the student was able to deduce that hydrogen bromide behaves as a catalyst in this two-step reaction.

(AQA Advanced Subsidiary GCE, Unit 2, May 2012, 1)

21 d) The concentration of hydrogen peroxide solution decreases. The rate of decomposition decreases.

e) i) $2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$

ii) Hydrogen bromide is regenerated in *Step 2*.

22 c) The time required will be longer.
In a larger container, concentrations of the reactants become smaller, and hence the rate of reaction decreases.

d) Colliding molecules will undergo reaction if they possess an energy equal to or greater than the activation energy, and collide in the right orientation.

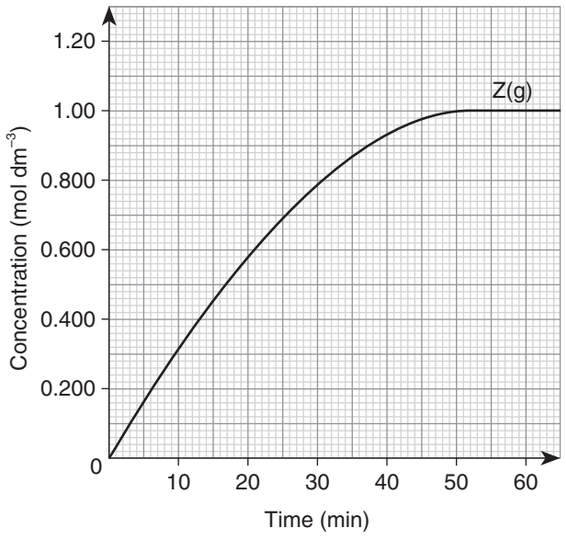
Unit Exercise

22 X(g) and Y(g) react according to the following equation:

$X(g) + Y(g) \longrightarrow 2Z(g)$

(X, Y and Z do not represent symbols of elements.)

Equal amounts of X(g) and Y(g) are allowed to react in a closed container of volume 1.00 dm^3 kept at a constant temperature. The graph below shows the change in concentration of Z(g) in the container with time.



a) From the graph, determine the instantaneous rate of formation of Z(g) at the 10.0 minute.
0.0292 mol dm^{-3} min^{-1}; refer to page T23.

b) Copy the graph and sketch the variation of the concentration of X(g) with time during the experiment.
Refer to page T23.

c) If the mixture of X(g) and Y(g) is allowed to react at the same temperature but in a closed container of volume 2.00 dm^3 instead, will the time required for the reaction to complete remain the same? Explain.
Refer to the previous page.

d) Explain why the collisions between molecules of X(g) and Y(g) will NOT necessarily lead to a reaction.
Refer to the previous page.

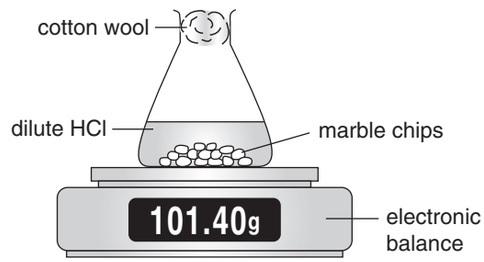
23 b) The rate of a reaction depends on how many effective collisions between reactant particles there are in a unit volume per unit time.
Only those colliding particles with an energy equal to or greater than the activation energy can react.
At a higher temperature, the reactant particles have more kinetic energy and collide more often.
More importantly, a larger portion of the particles have an energy equal to or greater than the activation energy.

23 A student investigated the rate of reaction between marble and hydrochloric acid. He used an excess of marble.

The reaction can be represented by this equation:

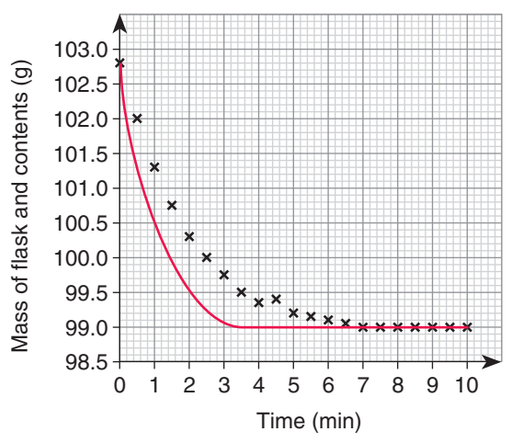
$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

The student used the apparatus shown in the diagram.



The student measured the mass of the flask and contents every half minute for ten minutes.

The results are shown on the graph. Use the graph to answer the questions.



The student repeated the experiment at a higher temperature. All other variables were kept the same as in the first experiment. The rate of reaction was much faster.

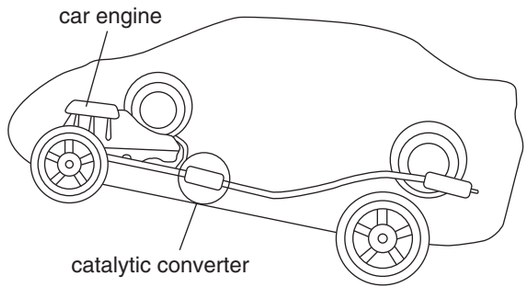
a) Draw a line on the graph to show what the results for this second experiment might look like.

b) Why does an increase in temperature increase the rate of reaction?

(AQA GCSE (Foundation Tier), Chemistry, Unit C2, Jan. 2008, 6(d))

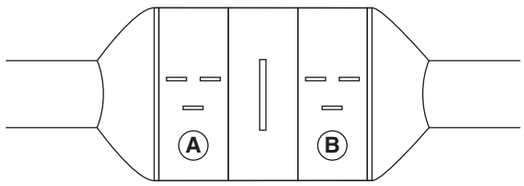
Unit Exercise

24 Read the information about car engines.

> Burning petrol in air is an exothermic reaction. This reaction is used in car engines.
>
> When petrol burns, it produces harmful substances such as nitrogen oxides and carbon monoxide.
>
> A catalytic converter stops these harmful substances being released into the air.
>
> 
>

a) The reaction is exothermic. What is the meaning of 'exothermic'?
Heat is released during the reaction.

b) The catalytic converter has two parts shown as A and B in the diagram.



Part A contains a catalyst made from platinum and rhodium.

Part B contains a catalyst made from platinum and palladium.

i) Why are catalysts used in chemical reactions?
Speed up the reaction.

ii) The equation for one of the reactions in part B is shown below.

Balance this equation.

$\underline{2}\,CO + O_2 \longrightarrow \underline{2}\,CO_2$

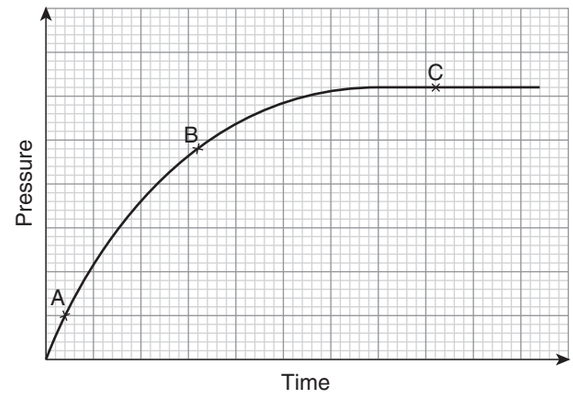
iii) The catalytic converter works for many years without replacing the catalyst.

Explain why the catalyst does not need to be replaced.
Catalyst is not used up in the reaction.

iv) Suggest why different catalysts are used in parts A and B. Different reactions require different catalysts.

(AQA GCSE (Higher Tier), Chemistry, Unit C2, Jan. 2011, 4(a), (b)(i), (iii)–(v))

25 In an experiment, a data-logger with pressure sensor was used to study the rate of decomposition of hydrogen peroxide (H_2O_2) in the presence of manganese(IV) oxide. The relation between the pressure and time measured is shown in the curve below.



a) The decomposition of hydrogen peroxide gives water and oxygen. After the experiment, it was found that the manganese(IV) oxide used did not undergo any chemical change.

i) State the function of manganese(IV) oxide.

ii) Explain why a pressure sensor could be used in this experiment.

iii) Write a chemical equation for the decomposition of hydrogen peroxide. Hence discuss the changes, if any, in the oxidation numbers of hydrogen and oxygen in the reaction.

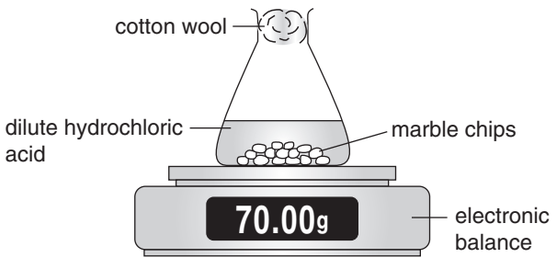
b) i) Explain why the respective rates of decomposition of hydrogen peroxide differ at points A, B and C on the curve.

ii) On the graph above, sketch a curve that should be obtained if the initial concentration of the hydrogen peroxide is HALF of its original value, while all other conditions remain unchanged.

(HKCEE, Paper 1, 2009, 10)

Unit Exercise

26 Andy is investigating the reaction between dilute hydrochloric acid and calcium carbonate (marble chips). He uses 50 cm^3 of the dilute acid and 5.00 g of marble chips.



When the reaction starts, fizzing can be seen as a gas is given off.

When the reaction stops, some marble chips can still be seen in the flask.

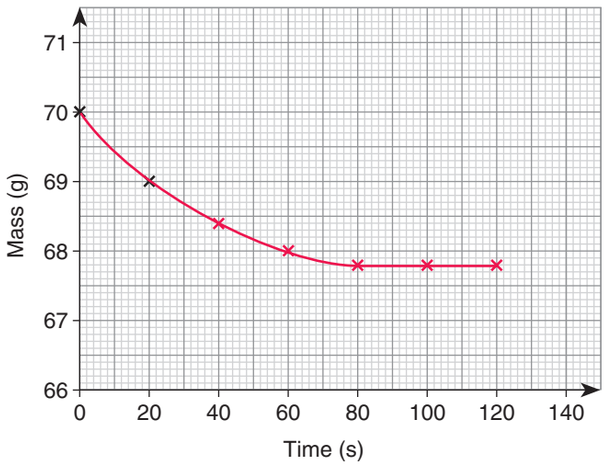
Andy measures the mass of the flask and its contents during the reaction.

His results are shown in the table below.

Time (s)	0	20	40	60	80	100	120
Mass (g)	70.00	69.00	68.40	68.00	67.80	67.80	67.80

a) i) Plot these results on this grid.

The first two points have been plotted for you.



ii) Draw the line of best fit.

b) Why has the mass decreased during the experiment?
Carbon dioxide gas escapes.

c) What is the purpose of the cotton wool in the neck of the flask?
To prevent spray / loss of the acid.

d) Explain why the last three masses in the table are the same. The acid has been used up.
No more gas is given off.

e) Andy wants to investigate the effect of using powdered marble chips.

He repeats the experiment with 50 cm^3 of the same dilute acid but using 5.00 g of powdered marble chips.

He notices that the mass lost is exactly the same but it happens in a much shorter time.

Why does the reaction take place more quickly than in the first experiment?
Powdered marble clips has a greater surface area.

f) Andy repeats the experiment again with 50 cm^3 of the same dilute acid and 10.00 g of powdered marble chips. The same amount of acid is used.
Marble clips are in excess.
Explain why the mass lost is still exactly the same as in the first two experiments.

(OCR GCSE 21st Century Science (Higher Tier), Additional Applied Science A, Unit A335/02, Jan. 2012, 4)

Unit Exercise